M.Sc. Thesis Project Summary

Prepared for Supervisor Meeting

Project Title

Comparative Codon Usage Bias Analysis in Drug-Sensitive versus Multidrug-Resistant *Mycobacterium tuberculosis*: Investigating Translational Selection and Mutational Pressure in Resistance-Associated Genes

Research Question

Do drug-resistant *M. tuberculosis* strains show measurably different synonymous codon usage patterns in resistance-associated genes compared to drug-sensitive strains, and if so, are those differences driven by random mutational pressure or by translational selection pressure?

Background and Rationale

Why Mycobacterium tuberculosis?

Tuberculosis (TB) kills approximately 1.3 million people per year globally. The emergence of multidrug-resistant TB (MDR-TB) — defined as resistance to both rifampicin and isoniazid, the two primary first-line antibiotics — is a major public health crisis. Understanding the genomic consequences of resistance acquisition beyond the known resistance mutations is an active and important area of research.

Why Codon Usage Bias?

Codon usage bias (CUB) refers to the unequal use of synonymous codons — different codons that encode the same amino acid. This is not random: organisms preferentially use codons that match the most abundant tRNA species in the cell, because translation is faster and more accurate when codon-tRNA pairing is optimized. Deviations from this optimization have measurable fitness costs.

Why Synonymous Mutations and Not Non-Synonymous?

Non-synonymous mutations in drug-resistant TB are extensively characterized — the specific mutations in rpoB, katG, and gyrA that cause resistance are well-documented in the literature. This is solved science. Studying them would be repeating known work.

Synonymous mutations change the codon but not the amino acid. They are traditionally assumed to be neutral, but this assumption is incorrect. A synonymous change from a common codon to a rare codon in a highly expressed gene slows translation and can affect protein folding. Whether resistance acquisition systematically shifts the synonymous codon landscape — and what evolutionary forces drive that shift — is much less studied. This is our research gap.

Data Source: The CRyPTIC Consortium

What is CRyPTIC?

CRyPTIC stands for **C**omprehensive **R**esistance Prediction for **T**uberculosis: an **I**nternational **C**onsortium. It is a large, multinational, peer-reviewed research initiative that collected M. tuberculosis clinical isolates from 23 countries worldwide, sequenced their genomes, and measured drug susceptibility for each isolate against 13 antitubercular drugs using standardized phenotypic drug susceptibility testing (DST).

Published Reference

The CRyPTIC Consortium. "A data compendium associating the genomes of 12,289 *Mycobacterium tuberculosis* isolates with quantitative resistance phenotypes to 13 antibiotics." *PLOS Biology* 20(8): e3001721 (2022). DOI: 10.1371/journal.pbio.3001721

Where the Data Lives

Raw Illumina FASTQ files are deposited in the European Nucleotide Archive (ENA). Accession numbers and per-isolate phenotypic metadata are provided in the CRyPTIC reuse table, publicly available at: https://ftp.ebi.ac.uk/pub/databases/cryptic/release_june2022/reuse/

Why This Dataset is Ideal

Feature	Detail
Sample size	12,289 isolates — 50 drug-sensitive + 50 MDR-TB selected
Phenotype confirmation	Each isolate has laboratory-confirmed DST result
Drug coverage	Binary R/S classification for 13 drugs including RIF and INH
Geographic diversity	23 countries — reduces geographic bias
Data format	Raw Illumina paired-end FASTQ — compatible with NGS pipeline
Peer-reviewed	Published in PLOS Biology — trusted, citable source

Methodology

Study Design

Two-group comparative genomic analysis: 50 drug-sensitive (DS-TB) isolates versus 50 multidrug-resistant (MDR-TB) isolates, selected from the CRyPTIC dataset based on confirmed phenotypic RIF_BINARY and INH_BINARY classifications.

NGS Pipeline

A reproducible Bash-based NGS pipeline processes raw reads through quality control, trimming, alignment, variant calling, and consensus sequence generation. Target genes (rpoB, katG, gyrA) are then extracted from each consensus sequence using Biopython. Reference genome: H37Rv (NC_000962.3).

Stage	Tool	Output
Download	prefetch + fasterq-dump	Raw FASTQ
Quality Control	FastQC + MultiQC	QC reports
Trimming	Trimmomatic PE	Trimmed paired FASTQ
Alignment	BWA MEM	SAM files
Sorting	samtools sort	Sorted BAM
Variant Calling	bcftools mpileup + call	VCF files
Consensus	bcftools consensus	Consensus FASTA + chain
CDS Extraction	Biopython + pyliftover	Gene sequences
Codon Analysis	Custom Python (RSCU, ENC)	Statistical output

Analytical Measures

RSCU — Relative Synonymous Codon Usage

RSCU measures which synonymous codons an organism actually prefers for each amino acid. An RSCU value of 1.0 means no preference — all synonymous codons used equally. Values above 1.0 indicate preference for that codon; below 1.0 indicates avoidance. Comparing RSCU profiles between DS-TB and MDR-TB groups shows whether resistance acquisition shifts codon preferences in target genes.

ENC — Effective Number of Codons

ENC quantifies the overall degree of codon bias. It ranges from 20 (maximum bias — only one codon used per amino acid) to 61 (no bias — all codons used equally). Critically, ENC can be plotted against GC content at third codon positions (GC3). If observed ENC values fall on the expected curve, mutational pressure is the main driver. If values deviate significantly below the curve, translational selection is at work — the organism is actively optimizing codon usage beyond what GC content alone would predict. This is the key test that distinguishes the two evolutionary forces.

Target Genes

Gene	Protein	Drug Resistance Link
rpoB	RNA polymerase beta subunit	Rifampicin resistance (S450L most common)
katG	Catalase-peroxidase	Isoniazid resistance (S315T most common)
gyrA	DNA gyrase subunit A	Fluoroquinolone resistance

Novelty and Contribution

Most TB genomics research focuses on identifying which non-synonymous mutations cause resistance — that question is largely answered. This project asks a different question: what happens to the synonymous codon landscape of resistance genes as strains evolve under antibiotic pressure? Specifically, does resistance acquisition impose a translational cost visible in codon usage patterns, and does the organism compensate through translational selection? This is a genome-level evolutionary question with implications for understanding the fitness cost of resistance and the long-term evolutionary trajectory of MDR-TB strains.

Data: CRyPTIC Consortium, PLOS Biology 2022 | Reference: H37Rv NC_000962.3 | Pipeline: BWA + samtools + bcftools + Biopython